

Project Development Phase

Date	9 July 2026
Team ID	SWTID-2026-5356
Project Name	India's Agricultural Crop Production Analysis
Maximum Marks	4 Marks

Data Preprocessing Steps

Step 1: Import the Dataset

- Imported the India's agricultural crop production analysis dataset into Tableau Public.
- Verified that all records and fields (State, District, Crop Year, Season, Crop, Area (Hectares), Production (Tone's), Yield, and other agricultural attributes) were imported successfully into the dataset.
- Confirmed that the dataset was ready for preprocessing.

Step 2: Check for Missing Values

- Examined the dataset for null or blank values.
- Verified that all records and fields (State, District, Crop Year, Season, Crop, Area (Hectares), Production (Tone's), Yield, and other agricultural attributes) were imported successfully into the dataset.
- Missing values were handled appropriately to ensure accurate analysis.

Step 3: Remove Duplicate Records

- Checked the dataset for duplicate agricultural records using State, District, Crop Year, Season, and Crop.
- Removed duplicate entries to prevent inaccurate production statistics and misleading analytical insights.
- Ensured that each agricultural production record was represented only once in the dataset.

Step 4: Validate Data Types

Ensured that each attribute had the correct data type.

For **India's Agricultural Crop Production Analysis**, use these data types:

- **State** → Geographic/String
- **District** → Geographic/String
- **Crop Year** → Number (Integer)
- **Season** → Categorical
- **Crop** → Categorical
- **Area (Hectares)** → Number (Decimal)
- **Production (Tonnes)** → Number (Decimal)
- **Yield** → Number (Decimal) (*Calculated: $Production \div Area$*)

Step 5: Verify Data Consistency

- Checked for invalid or unrealistic values such as negative Area, Production, or Yield, and corrected inconsistent records where necessary.
- Standardized State names, District names, Crop names, and Season values to maintain uniformity across the dataset.
- Ensured consistency across all categorical fields before performing data analysis and visualization.

Step 6: Prepare Calculated Fields

Created calculated fields required for dashboard analysis, including:

- Total Production
- Total Cultivated Area
- Average Yield (*$Production \div Area$*)
- Production by Crop
- Production by State
- Area by State
- Crop-wise Yield
- KPI Measures for Dashboard (*Total Production, Total Area, Average Yield, Number of States, and Number of Crops*)

Step 7: Final Clean Dataset

- Verified that the cleaned dataset was complete and suitable for analysis.
- Confirmed that all preprocessing activities were completed successfully.
- The dataset was finalized for dashboard creation and business intelligence visualizations in Tableau Public

Business Questions with Visualization

S.No.	Business Question	Visualization
1	How many agricultural production records are available for analysis after data preprocessing?	Total Records (KPI Card)
2	What is the total crop production across India?	Total Production (KPI Card)
3	Which crop has the highest production?	Crop-wise Production Analysis
4	How does crop production vary across different states?	State-wise Production Analysis (Map/Bar Chart)
5	Which states have the largest cultivated area?	State-wise Cultivated Area Analysis
6	How does average yield vary among different crops?	Crop-wise Yield Analysis
7	What is the seasonal distribution of crop production?	Season-wise Production Analysis
8	How has crop production changed over different crop years?	Year-wise Production Trend Analysis (Line Chart)

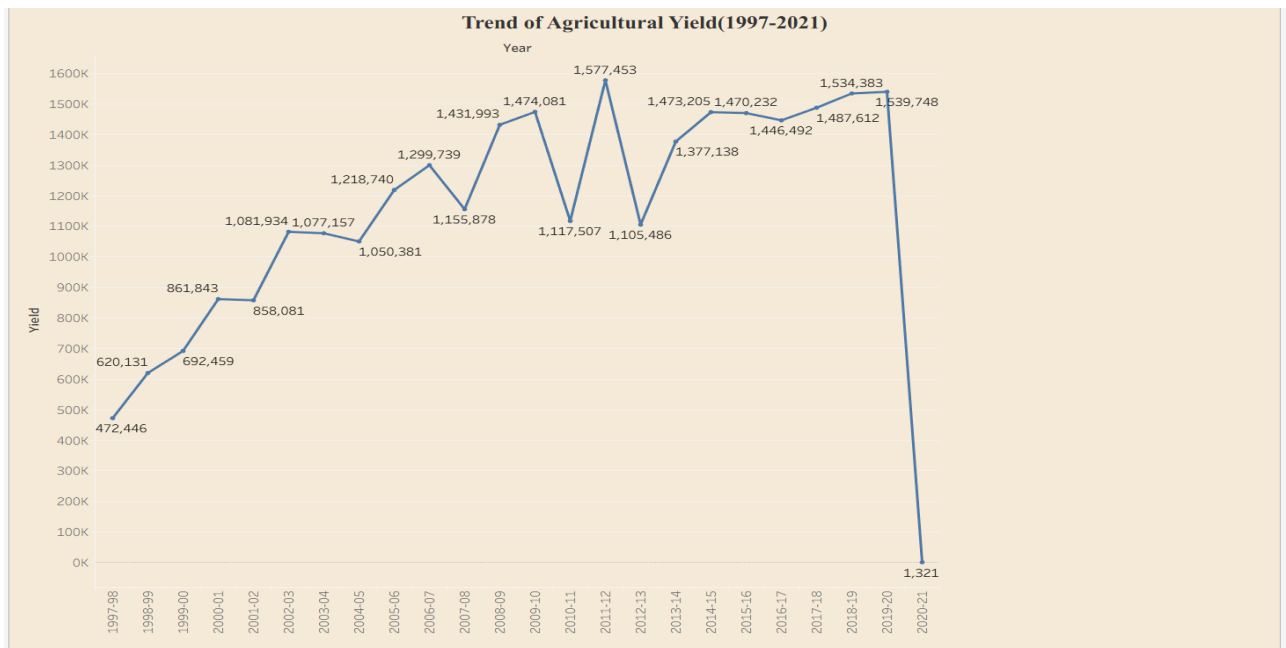
3.Visualiztions:

Total Production and Yield

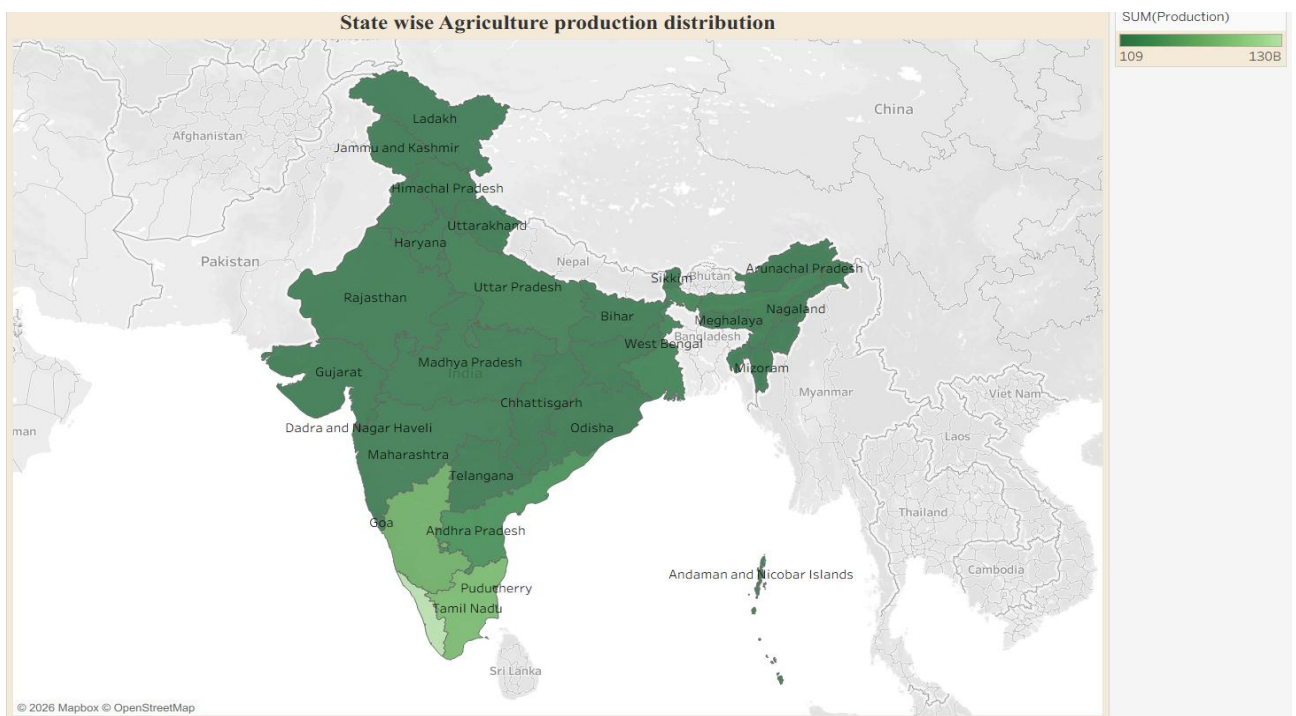
Production	330,997,839,211
Yield	27,425,439
Area	4,031,130,486

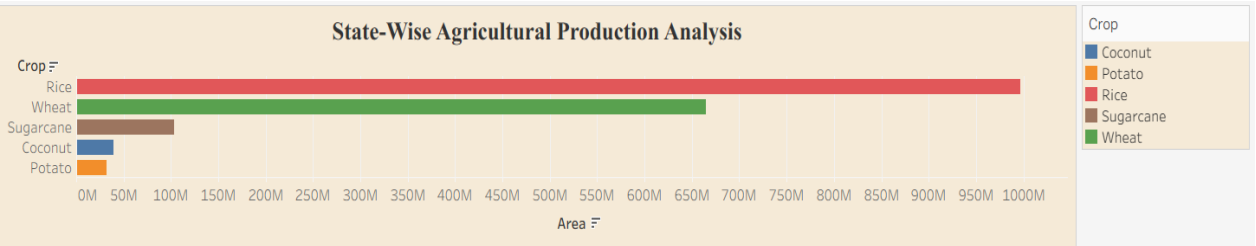
Total Production & Total Yield

Trend of Agricultural Yield (1997-2021)

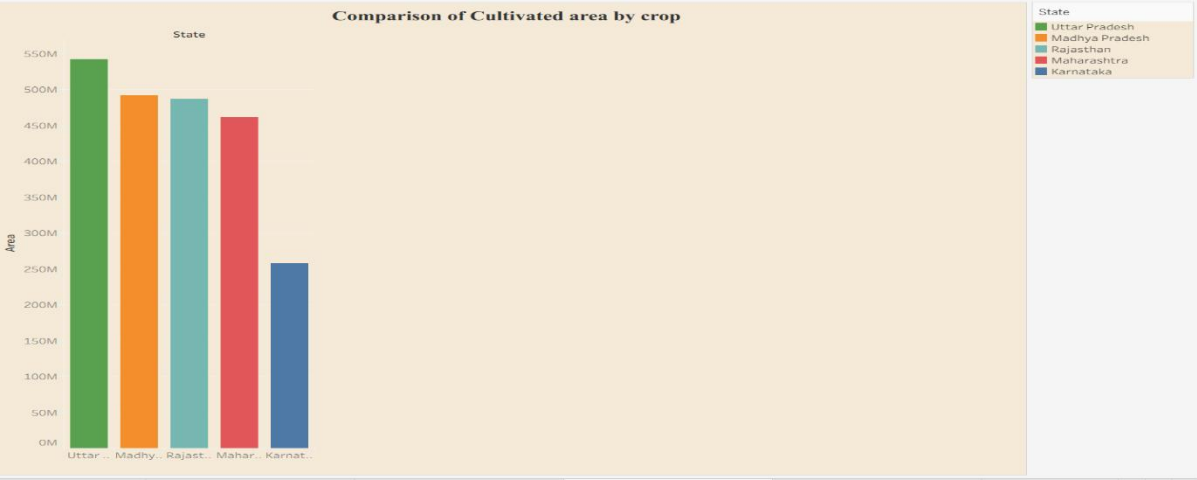


State Wise Agricultural Production distribution

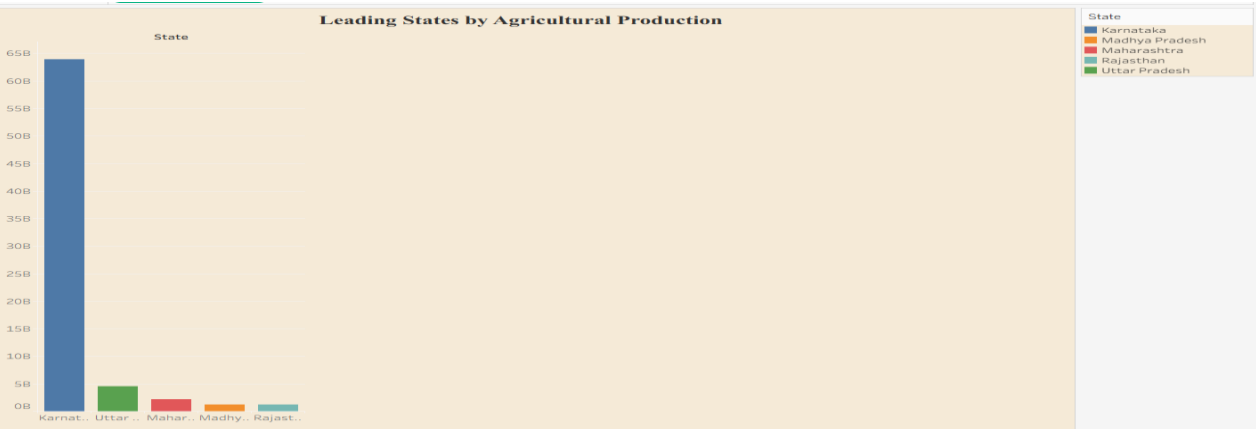




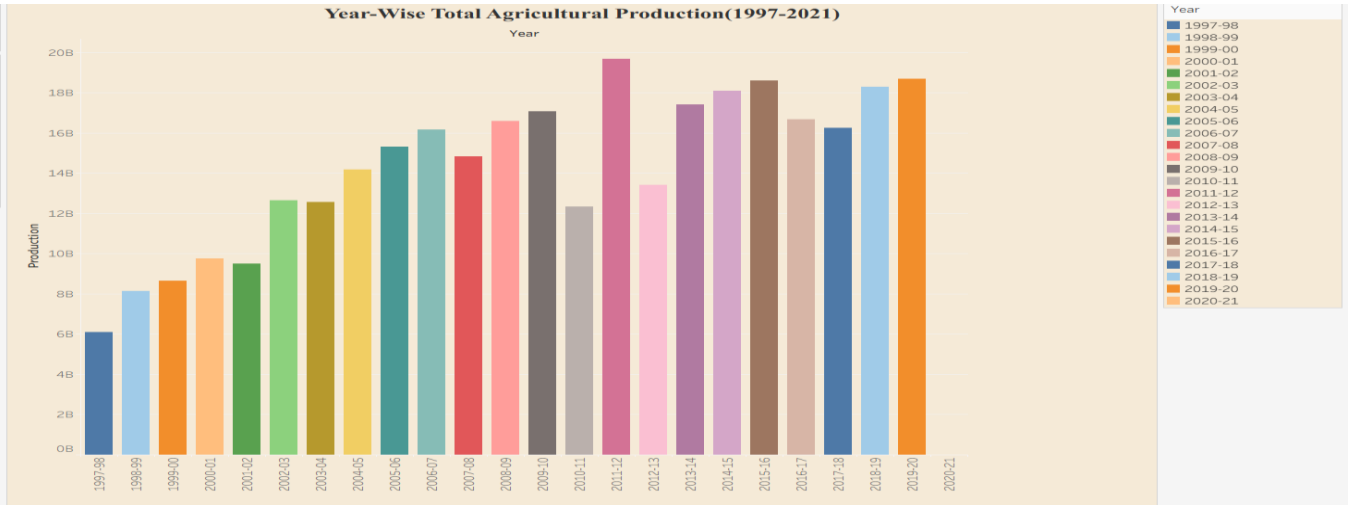
Comparison of cultivated area by crop



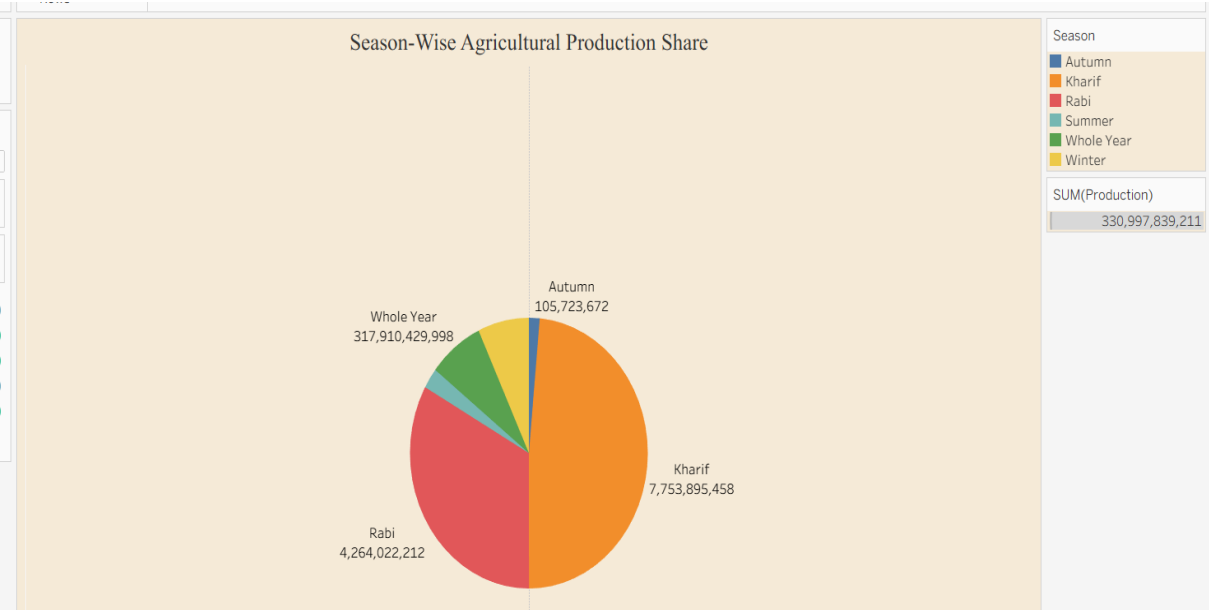
Leading states by Agricultural Production



Year-Wise Total Agricultural Production

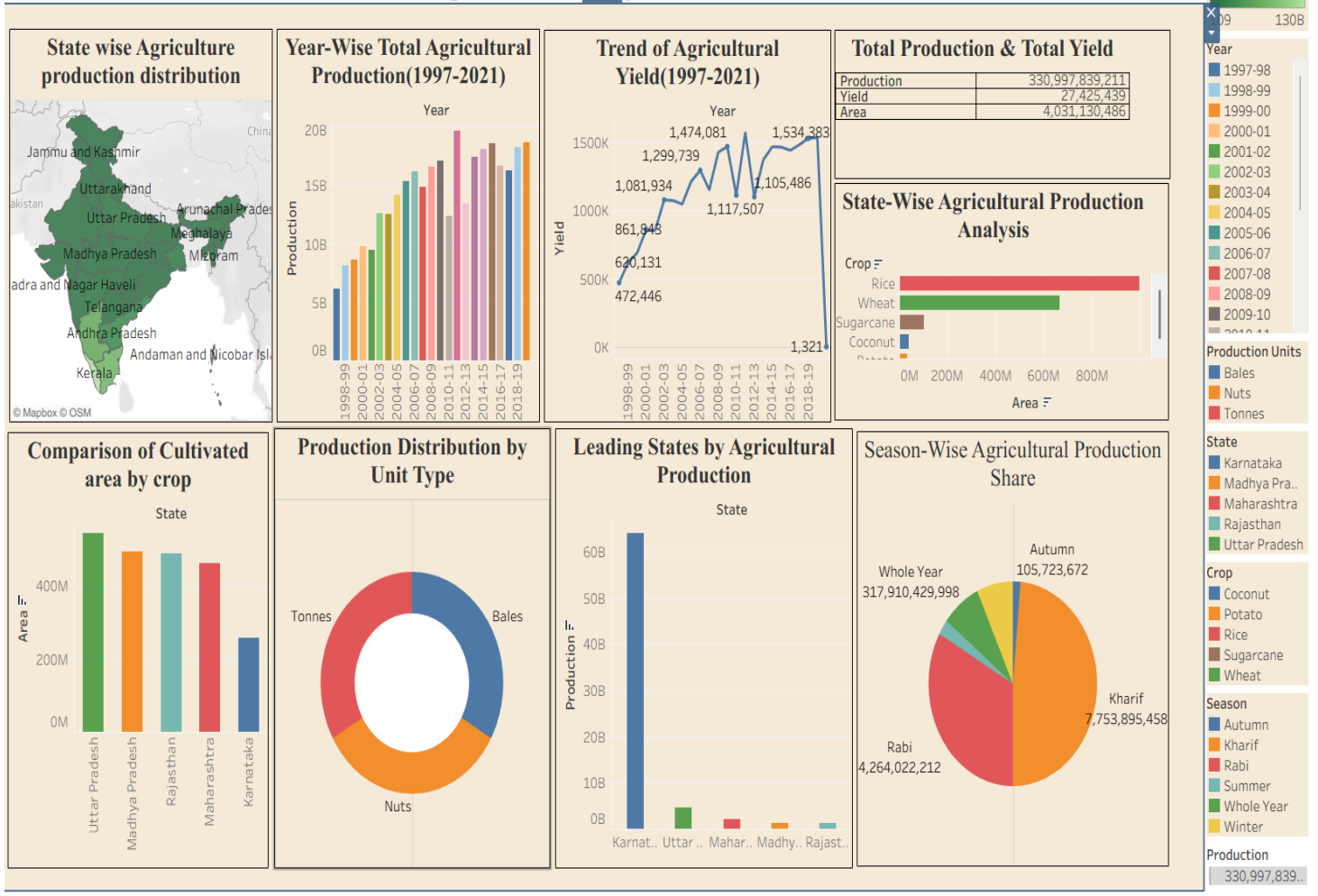


Season-Wise Agricultural Production Share



Dashboard:

Responsive and Design of Dashboard



Production: 330,997,839,211

Year: 1997-98, 1998-99, 1999-00, 2000-01, 2001-02, 2002-03, 2003-04, 2004-05, 2005-06, 2006-07, 2007-08, 2008-09, 2009-10, 2010-11, 2011-12, 2012-13, 2013-14, 2014-15, 2015-16, 2016-17, 2017-18, 2018-19

Production Units: Bales, Nuts, Tonnes

State: Karnataka, Madhya Pradesh, Maharashtra, Rajasthan, Uttar Pradesh

Crop: Coconut, Potato, Rice, Sugarcane, Wheat

Season: Autumn, Kharif, Rabi, Summer, Whole Year, Winter

Production: 330,997,839,211